Practice Exam - 1

Question 92 of 117

A 63-year-old with history of heart failure with preserved ejection fraction, type II diabetes mellitus, paroxysmal atrial fibrillation on anticoagulation, morbid obesity (BMI 42), and OSA intermittently adherent to CPAP presents to your clinic with worsening fatigue, progressive dyspnea, and exertional intolerance over the past 3 months.

Her vital signs are: BP 150/90 mmHg, HR 110 bpm, RR 22 breaths/min, SpO2 98% at room air.

Physical exam shows JVP 10 cmH2O, irregularly irregular rhythm, tachycardic, S3 and S4 summation gallop, and 2+ lower extremity edema. Echocardiogram shows left ventricular ejection fraction 60%, moderate concentric left ventricular hypertrophy, moderate left atrial enlargement, moderate tricuspid regurgitation, and estimated pulmonary artery systolic pressure of 35 mmHg. The patient is referred for invasive exercise hemodynamics.

Which of the following is the most likely finding with respect to exercise hemodynamics as compared to those of a normal individual?

A. Increase in PCWP from 12 to 16 mmHg

B. PCWP/cardiac output slope of 2.5 mmHg/L/min during exercise

C. Increase in pulmonary vascular resistance from 2.5 to 3 WU

D. Decrease in systolic blood pressure with exercise from 150 to 80 mmHg

Commentary | References

Testing Point: The board certified AHFTC specialist should be able to recognize important and prognostic exercise hemodynamic derangements in patients with heart failure with preserved ejection fraction.

Correct Answer: B. PCWP/cardiac output slope of 2.5 mmHg/L/min during exercise

Rationale: Compared with typical patients, physiologic changes in HFpEF include an elevated response in systemic vascular resistance and mean pulmonary artery (PA) pressure with exercise. Patients with HFpEF often display an increase in pulmonary capillary wedge pressure (PCWP) disproportionate to the increase in cardiac output (CO), as reflected in a PCWP/CO slope >2.

The other choices are incorrect.

Answer A: Although an increase in PCWP during exercise may occur in HFpEF, an isolated rise in PCWP is less specific than an abnormal PCWP/cardiac output slope, which better captures the disproportionate increase in filling pressures relative to cardiac output and has stronger diagnostic and prognostic value.

Answer C: The pulmonary vascular resistance would not be expected to increase with exercise unless the patient had concomitant HFpEF and pulmonary hypertension (PH) due to pre- and postcapillary components (CpcPH). The PVR reported here is not consistent with CpcPH.

Answer D: The dramatic decline in systolic blood pressure with exercise reported in this choice would be a poor prognostic sign but is not the typical or most common hemodynamic perturbation seen with exercise in HFpEF patients. Such a decline with exercise suggests a superimposed process, such as ischemia or dynamic outflow tract obstruction.